

Conveyor Jam Detector For Industries

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The conveyor jam detector circuit presented here is designed to activate an alarm device or programmable logic controller (PLC) when the speed of the conveyor system falls below the set minimum speed, alerting the workers the possibility of jamming of the conveyor system. Conveyor jam detector is an indispensable sensor device found in most conveyor safety and control systems.

While packages of various shapes and sizes move down a conveyor belt, it is imperative that they continue to travel down to prevent potential delays or severe damages due to an accidental jam. Following the common industrial practice, the sensor head of the conveyor jam detector can be installed in the second return idler support from the drive-end. (Normally when the speed of the conveyor belt is

slowing down, the rotation speed of the return idler is slowing down as well.) The conveyor jam sensor head installation is shown in Fig. 1.

Design description

The circuit diagram of the conveyor jam detector is shown in Fig. 2. It is built on a dual precision monostable multivibrator HCF4538B that offers stable retriggerable/resettable one-shot operation for any

fixed-voltage timing application. It is advisable to connect the entire electronic assembly to an industry-grade linear/switch mode DC power supply adaptor.

You can design a range of proximity sensors as the sensor head for the conveyor jam detector but using a readymade inductive (or optical) proximity sensor can save both time and effort.

The circuit diagram is very clear and self-explanatory. The first monostable (IC2A) of IC2, which comprises pins 1 through 7, is configured as a retriggerable monostable, and the leading edge of its output pin 6 is used to trigger the second monostable (IC2B), which comprises pins 9 through 15. LED1 and LED2 are system status indicators wired to the output of the first and second monostable, respectively. RC timing

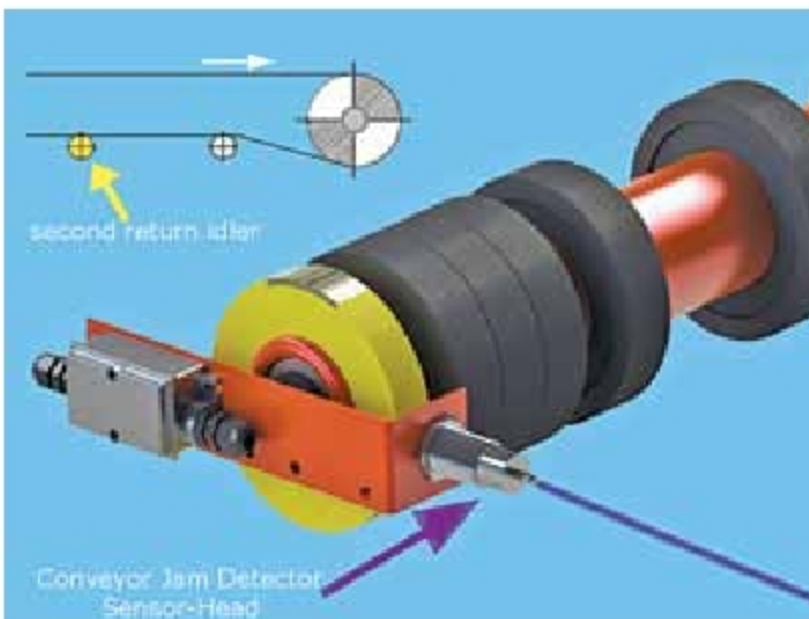


Fig. 1: Conveyor jam sensor head installation

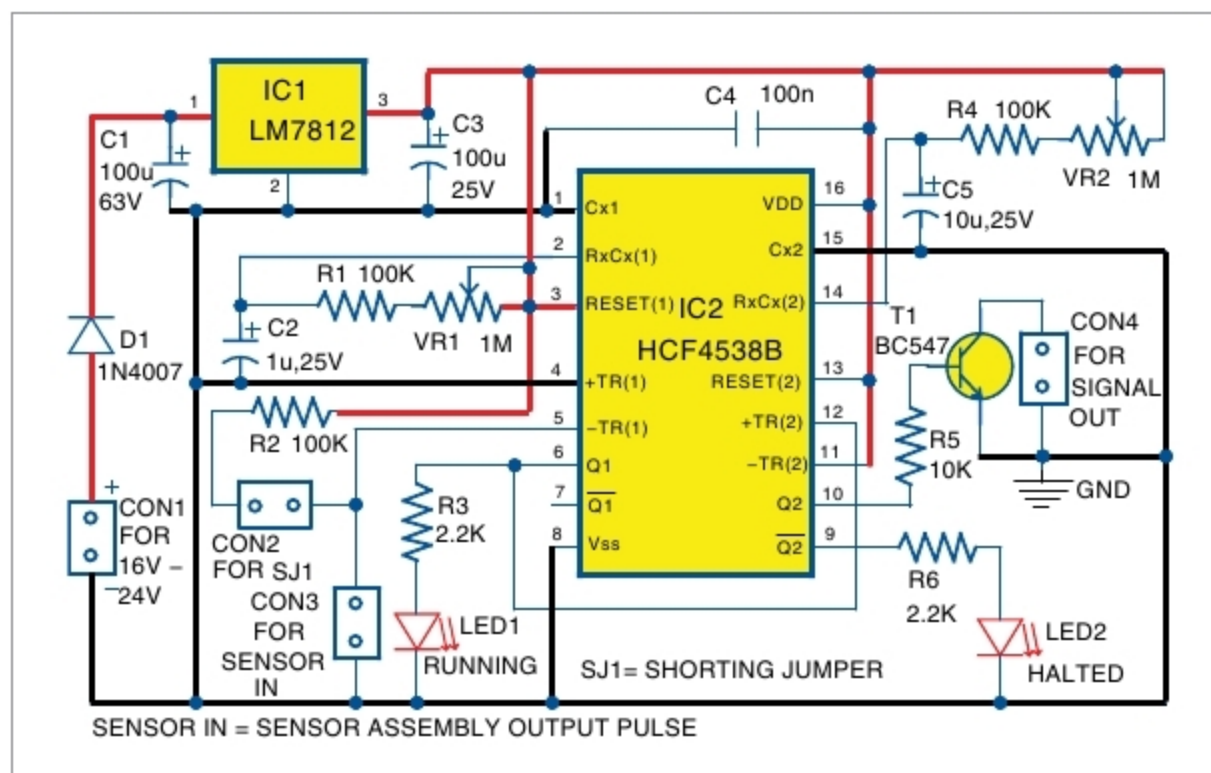


Fig. 2: Circuit diagram of conveyor jam detector

PARTS LIST

Semiconductors:

- IC1 - LM7812, 12V voltage regulator
- IC2 - HCF4538B dual monostable
- T1 - BC547 npn transistor
- LED1, LED2 - 5mm LED
- D1 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2, R4 - 100-kilo-ohm
- R3, R6 - 2.2-kilo-ohm
- R5 - 10-kilo-ohm
- VR1, VR2 - 1-mega-ohm potentiometer

Capacitors:

- C1 - 100 μ F, 63V electrolytic
- C2 - 1 μ F, 25V electrolytic
- C3 - 100 μ F, 25V electrolytic
- C4 - 100nF ceramic disk
- C5 - 10 μ F, 25V electrolytic

Miscellaneous:

- CON1 - 2-pin terminal connector
- CON2-CON4 - 2-pin connector
- 16V-24V unregulated power supply
- 16-pin IC base
- SN04-N inductive proximity sensor (optional)
- 4N25 opto-coupler (optional)

(All these are available from kitsnspares.com)